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AnalystLab Africa - Data Science Internship Programme

Week 4 Assignment

Machine Learning Problem Definition Report

**Improving Patient Appointment Attendance and Healthcare Support Using Data and
AI**

HealthConnect Clinic Experience Lab

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Executive Summary

HealthConnect Clinic is a fictional outpatient healthcare provider seeking to reduce missed appointments and improve patient support. This Week 4 study defines the Data Science problem and assesses whether the supplied historical appointment data can support a no-show prediction solution. It follows the early IBM Data Science Methodology stages: Business Understanding, Analytical Approach, Data Requirements, Data Collection and initial Data Understanding.

The dataset contains 5,000 appointment records, 18 variables and 1,696 anonymised patient identifiers. Outcomes comprise 2,423 no-shows (48.46%), 2,314 attended appointments (46.28%) and 263 cancellations (5.26%). The data can support a learning prototype, but missing values, operational conflicts, uncertain feature timing and inconsistent repeated patient histories prevent claims of production readiness.

Provisional decision: Use supervised binary classification for an initial attended-versus-no-show experiment. Preserve cancellations separately and use only information available before the appointment.

1. Project and Resource Review

Business scenario

HealthConnect Clinic manages appointment-based outpatient services. It reports patients missing appointments, difficulty understanding no-show factors, inefficient use of appointment slots and repetitive administrative enquiries. The central question is how the clinic can use data and AI to reduce missed appointments and improve patient support.

Data Science responsibility

The Data Science track must define the machine-learning problem and assess whether the available data can support no-show prediction. This includes proposing the target and features, deciding how cancellations will be handled, planning modelling and evaluation, and documenting factors that could affect validity or use.

Resources reviewed

Resource	Purpose
Week 4 assignment	Defines the scenario, Data Science tasks, scope and deliverables.
Appointment dataset	Provides 5,000 fictional, anonymised appointment records.
Data dictionary	Defines the intended meaning, type and notes for all 18 variables.
Clinic knowledge base	Provides operating context, appointment procedures and safety boundaries.

Initial approach

Step 1: Translate the clinic challenge into a precise and actionable Data Science question.

Step 2: Compare the analytical data requirements with the supplied resources.

Step 3: Inspect structure, completeness, consistency and operational plausibility.

Step 4: Define the target population, prediction time, target and candidate features.

Step 5: Plan modelling and evaluation without training a final model in Week 4.

Step 6: Record assumptions, limitations, risks and dependencies with each decision.

2. Business Understanding

Business problem and objective

Missed appointments may leave capacity unused, disrupt scheduling and limit organised patient support. HealthConnect currently has limited data-driven capability to identify appointments that may be at greater risk of becoming no-shows.

The objective is to improve appointment attendance and the use of scheduled capacity by determining whether historical information can support earlier identification of appointments that may benefit from additional attendance support.

Stakeholders and intended use

Stakeholder	Interest or responsibility
Clinic management	Operational efficiency, capacity utilisation and governance.
Administrative staff	Practical use of risk information for attendance support.
Healthcare professionals	More predictable schedules and appropriate use of clinical time.
Patients	Clear, fair and non-punitive appointment support.
Data and ML teams	Development, validation, implementation and monitoring.

A possible future action is to prioritise selected appointments for confirmation messages, reminder follow-ups or rescheduling assistance. This remains a proposed workflow because staff capacity, timing and escalation procedures have not been defined.

Scope and success

Week 4 covers problem and resource review, initial data assessment, target and feature definition, cancellation handling, modelling design and key considerations. It excludes final model training, deployment, clinical decision-making, causal claims about reminders and development of the healthcare information assistant.

Future success should be measured at two levels. Predictive performance should show whether no-shows can be identified with useful precision and recall. Business evaluation should later test whether a defined intervention improves attendance or appointment-slot use without unfairly burdening patients.

3. Analytical Approach

The project combines descriptive, diagnostic and predictive approaches. Description establishes what occurred; diagnosis explores variables associated with outcomes; prediction estimates future no-show risk. Prescriptive decisions about staff actions require separate operational design and evidence.

Element	Definition
Learning type	Supervised learning because historical outcomes are supplied.
Primary task	Binary classification for the initial attended-versus-no-show experiment.
Unit of analysis	One scheduled appointment record.
Proposed output	A no-show probability and an operationally chosen risk threshold.
Prediction point	After booking but before the scheduled appointment, at a clinic-defined time.
Analytical question	Can pre-appointment information predict attendance versus no-show?

Key distinction: The business problem is missed appointments; the Data Science problem is prediction; and the business action is the attendance-support process informed by the prediction.

4. Data Requirements and Collection

Required data

The analytical approach requires appointment-level history with a known outcome, a stable appointment identifier, a reliable prediction timestamp and predictors available before that timestamp. Relevant content includes scheduling details, booking lead time, prior attendance history, reminder activity, accessibility indicators and operational estimates.

Requirement	Expected representation
Outcome	A consistent categorical status: Attended, No-Show or Cancelled.
Dates	Parseable booking and appointment dates with a stated convention.
Numerical predictors	Valid units, documented ranges and explicit missing values.
Categorical predictors	Controlled categories with stable spelling and meaning.
Historical variables	Counts calculated strictly before the current appointment.
Availability	A rule or timestamp showing when each feature becomes available.

Collection assessment

The CSV and dictionary provide structured records and definitions. No external clinical or identifiable patient data is required for the educational prototype. However, reminder and cancellation timestamps, clinic location per appointment, intervention outcomes and an authoritative feature-availability map are absent. These gaps restrict operational interpretation.

5. Initial Data Assessment

Dataset overview

Measure	Observed value
Records	5,000 appointments
Variables	18 columns
Distinct patient identifiers	1,696
Appointment date range	1 January 2025 to 30 June 2026
Age range	18 to 80 years
Exact duplicate rows	0
Duplicate appointment identifiers	0

Outcome distribution

Outcome	Count	Share
No-Show	2,423	48.46%
Attended	2,314	46.28%
Cancelled	263	5.26%
Total	5,000	100.00%

Data-quality findings

Finding	Evidence	Implication or action
Missing distance	90 records (1.8%)	Investigate pattern; impute within training data or use a missing indicator.
Missing waiting time	60 records (1.2%)	Confirm pre-prediction availability before treatment.
Reminder channel None	1,366 records; all reminder_sent = No	Treat as a meaningful no-channel category.
Date format mismatch	Dictionary says ISO; CSV uses month/day/year	Parse explicitly and document the observed format.
Sunday conflict	737 Sunday appointments; stated clinic closure	Seek clarification; do not silently delete records.
Repeated-ID inconsistency	1,343 IDs span more than two years of age	Do not assume reliable patient histories without validation.

Suitability conclusion

The dataset is suitable for a controlled educational prototype because it contains a sizeable labelled outcome, a nearly balanced binary cohort and plausible predictors. It is not evidence of real clinical readiness because the data is fictional, important timestamps are absent, operational conflicts remain and repeated patient identifiers appear unreliable.

6. Target Variable and Cancellation Handling

The source target, `appointment_outcome`, has three categories. The proposed initial target is `no_show_flag`: No-Show = 1 and Attended = 0. Cancelled records will be preserved in the raw data but excluded from the first binary cohort.

Outcome	Initial treatment	Reason
No-Show	Target 1	Positive class of operational interest.
Attended	Target 0	Negative comparison class.
Cancelled	Excluded from initial binary cohort	Cancellation is neither attendance nor an unreported missed appointment.

The resulting cohort contains 4,737 appointments: 2,423 no-shows and 2,314 attended appointments. The no-show rate is 51.15%, so severe class imbalance is absent and synthetic oversampling is not justified initially.

This target answers a conditional question among records that ultimately did not cancel. Since future cancellations are unknown at booking, the population must be reconsidered before operational scoring. Later options include a three-outcome model, prediction after a cancellation window or a separate cancellation workflow.

7. Potential Input Features

Group	Candidate variables	Key consideration
Demographics	age; gender	Assess fairness and necessity; avoid duplicating <code>age_group</code> .
Appointment	type; day; time	Known when the appointment is scheduled.
Booking	<code>booking_lead_days</code>	Derived correctly and available after booking.
History	previous appointments; previous no-shows	Counts must precede the current appointment.
Reminder	sent; channel	Potential leakage if prediction occurs before reminder allocation.
Accessibility	distance to clinic	Limited missingness; calculation method unspecified.
Operations	waiting time	Availability timing is unclear and must be confirmed.

Excluded or deferred variables

- `appointment_id`: a record identifier rather than an explanatory feature.
- `patient_id`: excluded as a direct predictor because it risks memorisation and appears inconsistent.
- `appointment_outcome`: the source target and unavailable as an input.
- `age_group`: initially omitted when continuous age is retained.
- raw dates: use for controlled temporal features and splitting rather than memorising calendar positions.

8. Initial Modelling and Evaluation Approach

Step 1: Preserve the original data and work from a reproducible analysis copy.

Step 2: Parse dates, convert numerical fields and document missing-value treatment.

Step 3: Define the binary cohort without overwriting the raw outcome.

Step 4: Use appointment date for a future-oriented training and testing split.

Step 5: Fit all preprocessing only on training data to prevent leakage.

Step 6: Establish a simple majority or prevalence baseline.

Step 7: Train logistic regression first, then compare selected tree-based models if justified.

Step 8: Select a threshold using predictive performance and clinic workload.

Evaluation measures

Measure	Purpose
No-show recall	Proportion of actual no-shows identified for support.
No-show precision	Proportion of flagged appointments that become no-shows.
F1-score	Balances precision and recall.
Confusion matrix	Shows correct and incorrect classification counts.
ROC-AUC and PR-AUC	Assess ranking performance across thresholds.
Calibration	Tests whether probabilities correspond to observed risk.
Subgroup review	Examines material differences in error rates.

Accuracy will not be used alone. Flagging almost every appointment can create high recall but an impractical workload. The operating threshold must therefore reflect clinic capacity and the relative consequences of missed and unnecessary follow-ups.

9. Assumptions, Limitations, Risks and Dependencies

Type	Item	Response
Assumption	Outcomes and historical counts are consistently generated.	Validate definitions and state the assumption.
Assumption	Selected features exist before prediction.	Create a feature-availability map.
Limitation	The dataset is fictional and synthetic.	Treat results as a learning prototype.
Limitation	Reminder and cancellation timestamps are absent.	Keep scope provisional and request timing.
Limitation	Repeated patient histories are inconsistent.	Avoid patient-level memorisation.
Risk	Leakage may inflate performance.	Use pre-prediction inputs and train-only preprocessing.
Risk	Risk scores may stigmatise patients.	Use supportive interventions and review subgroup errors.
Risk	Association may be mistaken for causation.	Avoid causal explanations from predictive patterns.
Dependency	Staff need a response process and capacity.	Define timing, ownership and workload.
Dependency	Sunday scheduling conflict needs clarification.	Confirm whether records are errors or synthetic rules.

10. Proposed Week 5 Focus

Week 5 should move into structured Data Understanding and early Data Preparation. The immediate focus is exploratory analysis of outcome patterns, investigation of quality concerns, confirmation of prediction timing and construction of a leakage-safe modelling cohort.

- Explore outcomes across appointment, booking, history, reminder and accessibility variables.
- Investigate missing-value patterns and the Sunday scheduling conflict.
- Confirm the prediction point and feature availability.
- Create a separate prepared dataset and transformation log.
- Define the time-based evaluation strategy before fitting models.

Conclusion

Week 4 establishes a defensible foundation for HealthConnect's Data Science workstream. The data can support an educational no-show prediction prototype, but usefulness depends on careful target definition, cancellation handling, feature timing, data-quality investigation and ethical operational design. The next phase should deepen Data Understanding and prepare a reproducible modelling dataset while retaining the intended business action.